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Efficient Ozone Production within Indochina Biomass Burning Plumes: From Near-Source to Downwind Regions

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Zhiheng Liao¹, Fangzhou Li², Jielan Xie³ & Zhiqiang Ma¹✉

This study investigates O₃ production and chemistry within Indochina biomass burning plumes, using ozonesondes, satellite, and reanalysis data (2000–2024). Elevated terrain over the burning region produces a “terrain chimney” effect, lifting plumes into the free troposphere and confining downwind transport largely above the boundary layer. On average, Indochina biomass burning contributes ~30 ppbv of O₃ enhancement in the lower free troposphere (LFT, 2–5 km) over Hanoi and Hong Kong—nearly twice the increase reported for North American wildfires—indicating more efficient O₃ production. Satellite/reanalysis data reveal a NO_x-limited O₃ formation regime over the Indochina Peninsula and along smoke transport pathways. On peak-fire days, NO₂ surges while CH₂O increases modestly, distinct from North America. Despite the NO_x surge, the O₃ formation regime remains NO_x-limited across most transport pathways, favoring sustained O₃ production via NO_x → O₃. Combined with strong solar radiation and high VOC background, this yields an average O₃ production rate of 17.6 ppbv h⁻¹ over northern Laos, and a dilution-corrected excess LFT O₃ concentration of up to 150 ppbv over the Greater Bay Area on smoke-laden days. Our findings underscore the urgency of biomass burning prevention in Indochina.

Tropospheric ozone (O₃) is a key atmospheric component influencing climate, air quality, and atmospheric chemistry¹. It is primarily formed through sunlight-driven photochemical reactions between volatile organic compounds (VOCs) and nitrogen oxides (NO_x). These O₃ precursors originate from both anthropogenic emissions, such as fossil fuel combustion, and natural sources, including biomass burning, lightning and biogenic processes. Open biomass burning—comprising prescribed fires, agricultural burning, and wildfires—is estimated to contribute approximately 170 Tg of O₃ annually on a global scale, accounting for about 3.5% of total tropospheric O₃ production². In the remote troposphere, however, O₃ enhancements from biomass burning emissions can be equal or even surpass those associated with urban emissions³.

Typical biomass burning emissions can enhance tropospheric O₃ levels not only within the burning areas but also in downwind regions, through long-range transport and chemical reactions occurring along the transport pathway^{4–9}. The in-situ measurements from Wildfire Experiment for Cloud chemistry, Aerosol absorption, and Nitrogen (WE-CAN) and Fire Influence on Regional to Global Environments and Air Quality experiment (FIREX-AQ) campaigns identified several critical factors governing O₃ production

from wildfires in the western United States, and highlighted rapid shifts in chemical regimes that profoundly affect O₃ formation chemistry during smoke transport^{10–12}. Full-coverage satellite observations of O₃ precursors from California wildfires indicated that nitrogen dioxide (NO₂) enhancements within fire plumes are largely confined to the immediate vicinity of the burning areas, whereas increases in formaldehyde (CH₂O, a proxy for VOCs) extend over much greater distances, reaching from the source regions to downwind urban locations¹³. As a result, NO_x-limited source regions experience a marked rise in NO_x, while downwind VOC-limited urban areas witness a pronounced increase in CH₂O, both of which enhance O₃ production efficiency. The 2019 Williams Flats wildfire event in the northwestern United States also exhibited a comparable chemical mechanism that promotes O₃ formation¹⁴. Nevertheless, given worldwide differences in vegetation types, topographic features, and chemical environments, whether other fire-prone regions share similar O₃ precursor chemistry remains uncertain and warrant in-depth investigation.

The Indochina Peninsula (ICP), encompassing Myanmar, Thailand, Laos, Cambodia, and Vietnam, is one of the world's most active biomass burning regions, with over 100,000 fire events occurring annually, the

¹Institute of Urban Meteorology, China Meteorological Administration, Beijing, China. ²Laboratoire des Sciences du Climat et de l'Environnement, LSCE, Gif-sur-Yvette, France. ³Guangdong Provincial Key Laboratory of Marine Disaster Prediction and Prevention, Institute of Marine Sciences, Shantou University, Shantou, China. ✉e-mail: zqma@ium.cn

majority of which take place during the spring fire season (March–April)^{15–17}. Previous studies have reported substantial increases in tropospheric O₃ concentrations within biomass burning plumes originating from the ICP^{5,6,18–22}, and this increase has been steadily amplifying over the long term, driven by the eastward shift of biomass burning activity within the ICP⁶. In these studies, the “fire-on” and “fire-off” simulations were typically employed to quantify the overall contribution of Indochina biomass burning to tropospheric O₃ enhancement^{18–21}, yet they often overlooked the underlying photochemical conditions that determine which precursor (NO_x or VOCs) predominates in the ozone formation pathway²³. Although several aircraft-based studies have reported the O₃ and its precursor concentrations within Indochina biomass burning plumes over the remote western Pacific^{21,24}, the O₃ chemistry along the complete transport pathway remains poorly understood, particularly in contrast to North America, where a relatively rich body of observational knowledge exists^{11,13,14}. Unlike the coniferous or mixed forests typical of extra-tropical North America, the vegetation in the tropical Indochina region is dominated by broadleaf forests²⁵. Consequently, the emissions of O₃ precursors from Indochina biomass burning likely differ from those in extra-tropical North America²⁶, potentially leading to different O₃-precursor chemistry. The issue is further complicated by region-specific interactions between biomass burning emissions and anthropogenic emissions. Overall, there is a pressing need to elucidate the O₃-precursor chemistry along the complete transport pathway of Indochina biomass burning plumes.

This study aims to enhance our understanding of O₃ production in springtime biomass burning plumes originating from the Indochina Peninsula. By integrating multi-site ozonesonde observations (Hanoi, Hong Kong, and Taipei, including Pengchiayu), multi-satellite remote sensing data, and meteorological and chemical reanalysis products over the period 2000–2024, we systematically quantify tropospheric O₃ enhancements attributable to Indochina biomass burning, evaluate the sensitivity of O₃ production to precursor species, and identify the dominant chemical pathways driving O₃ build-up within these plumes along a gradient from near-source to downwind regions. In addition, we compare our findings with previous results from extra-tropical North America to highlight the principal similarities and differences between the two regions.

Results and discussion

Substantial enhancement of O₃ within Indochina biomass burning plumes

The multi-site O₃ climatology (Figure. S1) shows marked O₃ maxima in the lower free troposphere (LFT, 2–5 km above sea level) over Hanoi and Hong Kong during the Indochina fire season (March–April). On average, the fire-season LFT O₃ concentrations exceed those in adjacent months by up to 18.0 ppbv (Hanoi), 20.0 ppbv (Hong Kong), and 10.0 ppbv (Taipei), all significant at the $p = 0.05$ level. In contrast, O₃ concentrations exhibit no significant variation in the near-surface boundary layer (below 500 m) between the fire season and adjacent months. Since biomass burning also occurs sporadically during adjacent months, this month-scale comparison only offers a rough estimate of tropospheric O₃ enhancement attributable to Indochina biomass burning emissions. Taking advantage of the abundant ozonesonde observations available in Hong Kong, we categorize these days with LFT O₃ concentration below the 25th percentile of all fire-season values as smoke-free days, while those with LFT O₃ concentration above the 75th percentile as smoke-laden days (Fig. 1, upper panel). Each category contains 50 cases, which are relatively evenly distributed across the study period. An analysis of trajectory-fire interception²⁷ (TFI; Fig. 1, middle panels) shows that the median TFI value on smoke-laden days (738) is seven times larger than that on smoke-free days (104). This finding suggests that the receptor-oriented classification enables robust source attribution to Indochina biomass burning emissions. Furthermore, in the core burning region of northern Laos, a pronounced fire activity peak (1,044 hotspots) occurs two days prior to smoke-laden days, whereas a fire activity minimum (105 hotspots) emerges two days before smoke-free days (Figure. S2). This temporal offset is consistent with the two-day transport timescale for

biomass burning plumes traveling from the Indochina Peninsula to Hong Kong. Accordingly, from a source perspective, we define the days preceding smoke-free and smoke-laden days by two days as rare-fire days and peak-fire days, respectively. The classification enables a daily-scale composite comparison, thereby yielding a more accurate quantification of the contribution of Indochina biomass burning emissions to tropospheric O₃ enhancement in downwind regions.

The lower panels of Fig. 1 show tropospheric O₃ profiles on smoke-free and smoke-laden days at Hanoi, Hong Kong, and Taipei. It is worth noting that the initial classification of smoke-free and smoke-laden days was determined from ozonesonde data only at Hong Kong. Given the limited number of ozonesonde observations at Hanoi and Taipei, as well as their proximity to the Indochina biomass burning source, we supplemented the labeled-day sets for Hanoi and Taipei by including the preceding day (for Hanoi) and the following day (for Taipei) relative to the Hong Kong-classified dates. Ultimately, Hanoi had 6 smoke-free days and 3 smoke-laden days, while Taipei had 3 smoke-free days and 5 smoke-laden days. On smoke-laden days, the average LFT O₃ concentrations at Hanoi and Hong Kong reached 93.4 and 84.7 ppbv, which were 28.9 and 32.2 ppbv (45% and 61%) higher than those on smoke-free days, respectively. In contrast, the average LFT O₃ concentrations at Taipei on smoke-laden and smoke-free days were relatively close, at 66.3 and 57.2 ppbv, respectively. Consistent with the earlier comparison between fire season and adjacent months, O₃ concentrations in the near-surface boundary layer showed no statistically significant differences between smoke-free and smoke-laden days. These results demonstrate that the influence of Indochina biomass burning on O₃ in downstream low-elevation regions is largely confined to the lower free troposphere. In a previous ozonesonde-based study, Moeini et al.²⁸ estimated wildfire-generated O₃ over North America and reported an average increase of approximately 15 ppbv in lower tropospheric O₃ concentrations from non-affected to fire-affected parcels. By comparison, the LFT O₃ enhancements observed at Hanoi and Hong Kong (28.9 and 32.2 ppbv, respectively) are nearly twice that in Northern America, implying more efficient O₃ production in Indochina biomass burning plumes than in their North American counterparts. This difference is likely driven not only by the stronger solar radiation owing to the tropical location of the Indochina Peninsula, but also by more favorable background precursor levels and chemical conditions for O₃ formation—points that will be further substantiated in the following sections.

In contrast to multi-site observations, the gridded EAC4 chemical reanalysis can illustrate a complete three-dimensional structure of tropospheric O₃ transport from near-source to downwind regions. Using fire-season ozonesonde data, we first demonstrate that the EAC4 reanalysis effectively captures tropospheric O₃ variability across our study region (Figure. S3). A further comparison of tropospheric O₃ enhancements on smoke-laden days reveals that the EAC4 reanalysis is in good agreement with the sonde-based observations (Figure. S4), thereby supporting its use for broader spatial analysis. Figure 2 presents the EAC4-based three-dimensional O₃ enhancements on the peak-fire and smoke-laden days relative to rare-fire days. The vertical cross-sections of tropospheric O₃ enhancements from northern Laos to Taipei are shown in the left panels of Figure. S5. In these figures, we observed a pronouncedly high O₃ enhancement (> 60 ppbv) in the atmospheric boundary layer of northern Laos, and a distinct high-O₃ transport corridor between 850 and 500 hPa extending from northern Indochina to the Beibu Gulf, and further to coastal southern China. Within this transport corridor, peak O₃ enhancements at the 700 hPa level reached up to 30 ppbv (> 60%), consistent with ozonesonde observations at Hanoi and Hong Kong (Fig. 1d, e). Under this elevated transport regime, surface layers in low-lying downwind regions remained largely unaffected, showing no significant O₃ variability between rare-fire and peak-fire (or smoke-laden) days. These findings align with previous model results in Hanoi²⁹ and observational statistics in Hong Kong⁶. As shown by the cross-sectional distribution, the elevated topography over the core burning region produces a pronounced “terrain-chimney” effect, which rapidly lifts biomass burning plumes into the free

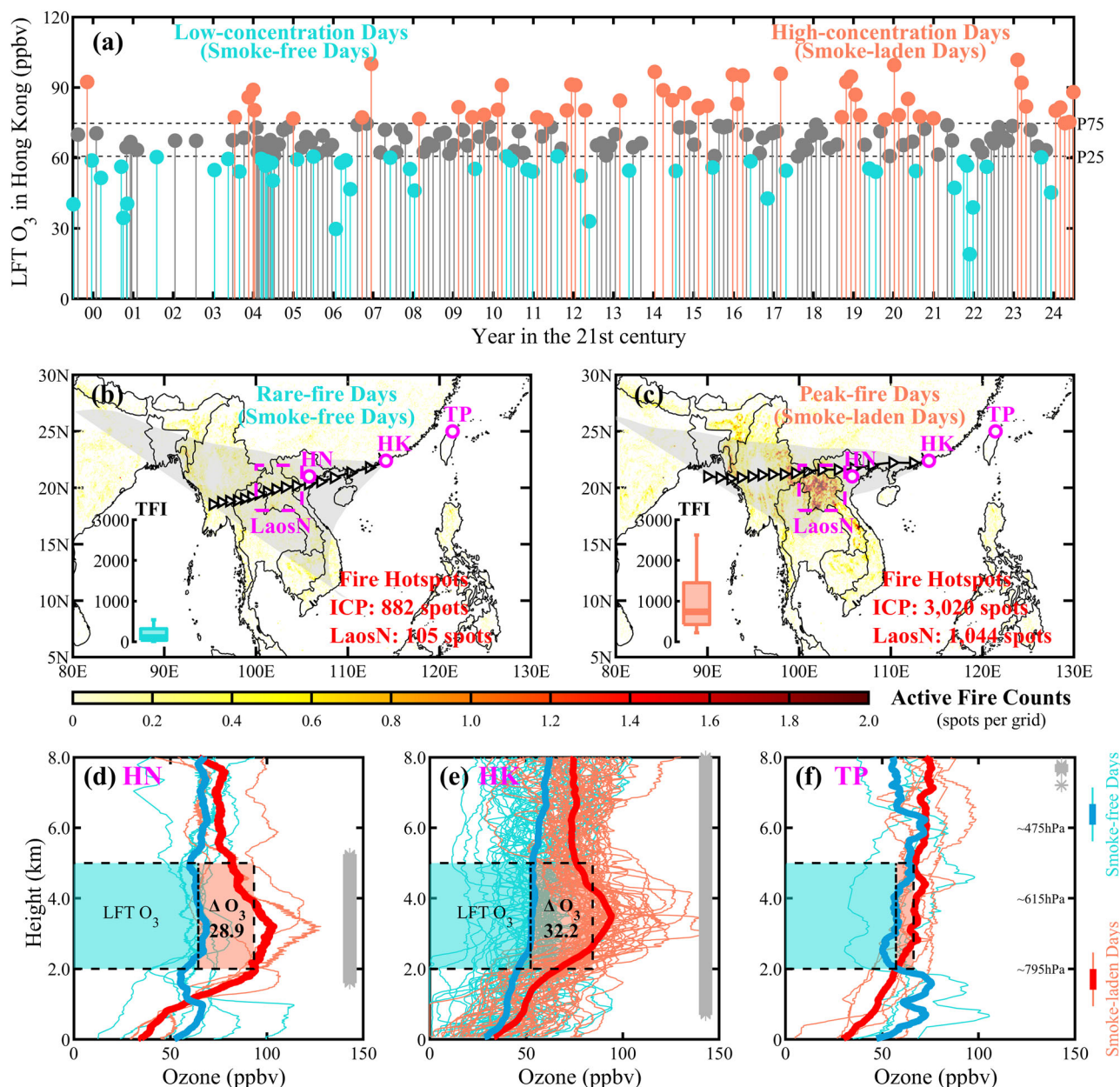


Fig. 1 | Relationships between sonde-based O_3 enhancements and Indochina biomass burning. **a** Daily LFT O_3 at Hong Kong during Indochina fire season, with smoke-free and smoke-laden days marked. **b, c** 96-h backward trajectories entering the LFT over Hong Kong on smoke-free and smoke-laden days (triangle lines: mean; gray shade: standard deviation) and the corresponding trajectory-fire interception (TFI) values (boxplots showing 5th, 25th, 50th, 75th, 95th percentiles), along with active fire distribution on rare-fire and peak-fire days. Red digits indicate the total number of fire hotspots in the Indochina Peninsula (ICP) and the core burning region of

northern Laos (LaosN; magenta dashed boxes). Magenta circles denote Hanoi (HN), Hong Kong (HK), and Taipei (TP). **d–f** Mean (bold lines) and individual (thin lines) O_3 profiles on smoke-free and smoke-laden days at Hanoi (HN), Hong Kong (HK), and Taipei (TP). Gray asterisks mark heights at which the difference of O_3 between smoke-laden and smoke-free days are statistically significant at the 95% confidence level. Shaded bars represent the mean LFT O_3 concentrations on smoke-free days and the excess O_3 concentrations (ΔO_3) on smoke-laden days relative to smoke-free days.

troposphere and confines their subsequent downstream transport largely within the lower free troposphere (Figure. S5). On smoke-laden days, high O_3 enhancements at 700 hPa were observed farther east compared to two days earlier (i.e., peak-fire days), suggesting additional O_3 formation and favorable O_3 -precursor chemistry during smoke transport. Further evidence for this in-transit O_3 formation comes from ozonesonde observations: although Hanoi, as a near-source site, recorded a higher LFT O_3 concentration on smoke-laden days than Hong Kong (93.4 ppbv vs. 84.7 ppbv), its excess relative to smoke-free days was actually slightly smaller (28.9 ppbv vs. 32.2 ppbv at Hong Kong). Furthermore, multi-site comparisons between

the fire season and adjacent months provide similar supporting evidence (Figure. S1; lower panels).

Substantial enhancement of O_3 precursors within Indochina biomass burning plumes

Based on the EAC4 Global Fire Assimilation System, we calculated the daily averaged emissions of NO_x and non-methane hydrocarbons (NMHC) from open biomass burning on rare-fire and peak-fire days (Figure. S6). Regardless of whether it is rare-fire or peak-fire days, biomass burning emissions in the Indochina Peninsula are substantially higher than those in

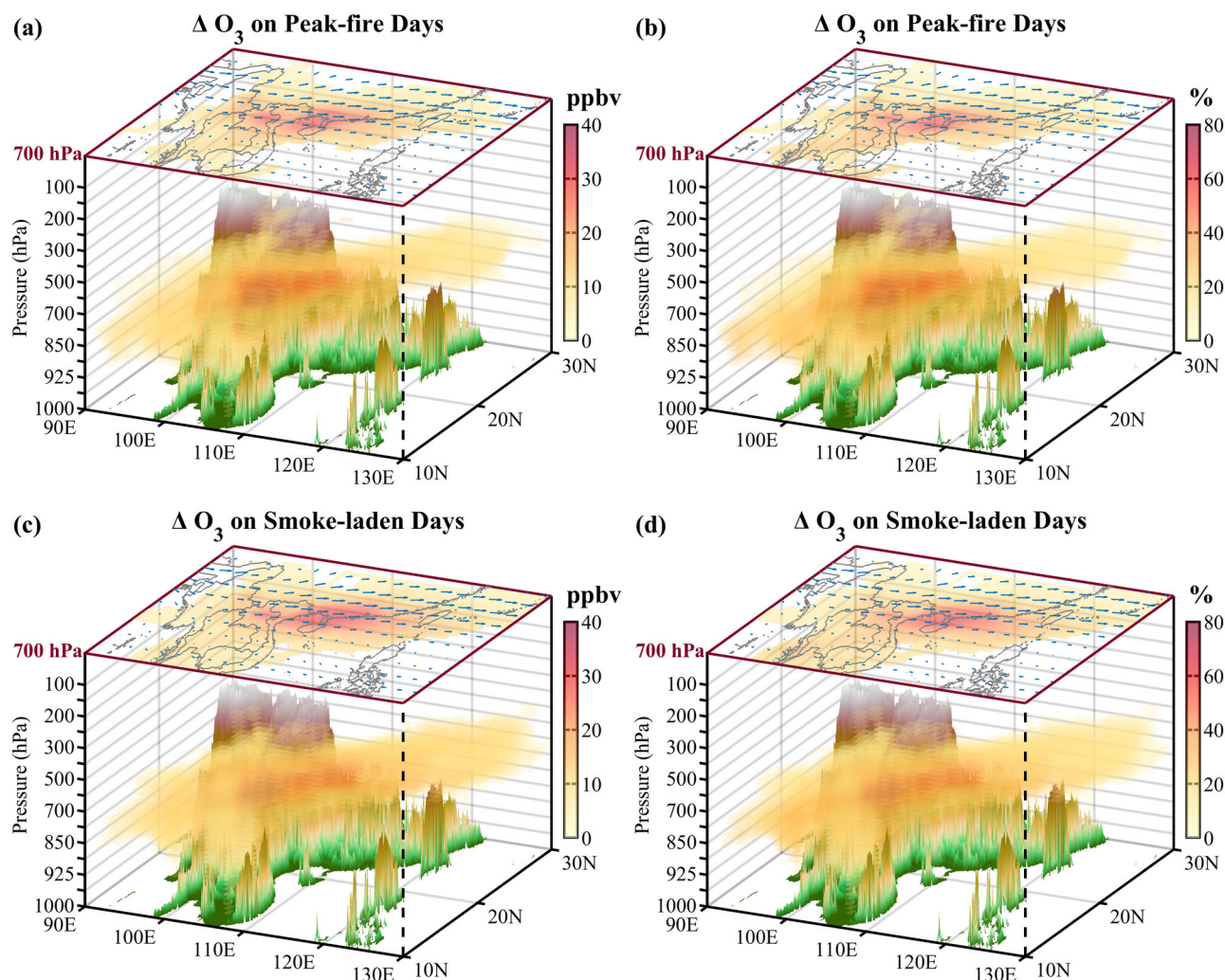


Fig. 2 | EAC4-based excess O_3 concentrations (ΔO_3) relative to rare-fire days. Upper panels **a, b** represent ΔO_3 on peak-fire days, while lower panels **c, d** display ΔO_3 on smoke-laden days. Left panels **a, c** show the absolute excess (ppbv), and right

panels **b, d** represent the percentage excess (%). Only areas with statistically significant increases at the 95% confidence level are shown. Red-bordered contours highlight the excess O_3 concentrations and the fire-season wind fields at 700 hPa.

surrounding areas, making emission changes outside the peninsula almost negligible. From rare-fire to peak-fire days, NO_x and NMHC emissions in the Indochina Peninsula increased approximately 2.8-fold, rising from 2.55 Gg day⁻¹ to 7.12 Gg day⁻¹ and from 7.71 Gg day⁻¹ to 21.81 Gg day⁻¹, respectively. Particularly, the core burning region of northern Laos experienced the most significant increase in emissions by approximately 4.0 times. Consequently, northern Laos's share of total biomass burning emissions in the Indochina Peninsula increased from about 31% on rare-fire days to 44% on peak-fire days. Evidently, the O_3 precursors released from open biomass burning in the Indochina Peninsula cause the observed increase in tropospheric O_3 concentration. However, the specific pathway of O_3 photochemical formation depends on the actual incremental ratios of these two-type precursors in the atmosphere and their relative ratio, which jointly determine the ozone formation chemistry.

Figure 3 shows the percentage increase of NO_2 and CH_2O on peak-fire and smoke-laden days relative to rare-fire days, reflecting the actual incremental ratios of NO_x and VOCs in the atmosphere due to biomass burning emissions in the Indochina Peninsula. Although Indochina biomass burning emits larger absolute amounts of NMHC than NO_x , satellite retrievals reveal that the percentage increase in tropospheric NO_2 column over the core burning region of northern Laos is substantially greater than that of CH_2O . The contrasting response between tropospheric CH_2O and NO_2 columns differs markedly from that observed over North America,

where significant NO_2 enhancements within fire plumes are largely confined to active burning regions, whereas CH_2O enhancements extend far downwind from the source areas to urban regions^{13,14}. Such pronounced regional differences likely stem from divergent background concentrations and vegetation types between northern Indochina and North America. In semi-arid, extra-tropical North America, the sparse vegetation results in low biogenic VOC emissions, leading to a low atmospheric VOC background concentration, while relatively high anthropogenic emissions contribute to a high NO_2 background. In contrast, the mountainous areas of tropical northern Indochina feature dense vegetation. Higher biogenic VOC emissions in northern Indochina contribute to a higher atmospheric VOC background, while lower anthropogenic emissions result in a lower NO_2 background concentration. This contrasting precursor background pattern is corroborated by global satellite observations, which show that tropospheric NO_2 columns over Indochina are lower than those over North America³⁰, whereas CH_2O columns are substantially higher over Indochina³¹. Consequently, the incremental contributions of the two types of O_3 precursors from biomass burning emissions to the corresponding atmospheric species differ markedly between these two regions.

The long-range transport of biomass burning plumes from the Indochina Peninsula occurs predominantly within the lower free troposphere (Figs. 1 and 2), implying that the high anthropogenic emissions in low-lying downwind regions (i.e., Hanoi and the Greater Bay Area) can substantially

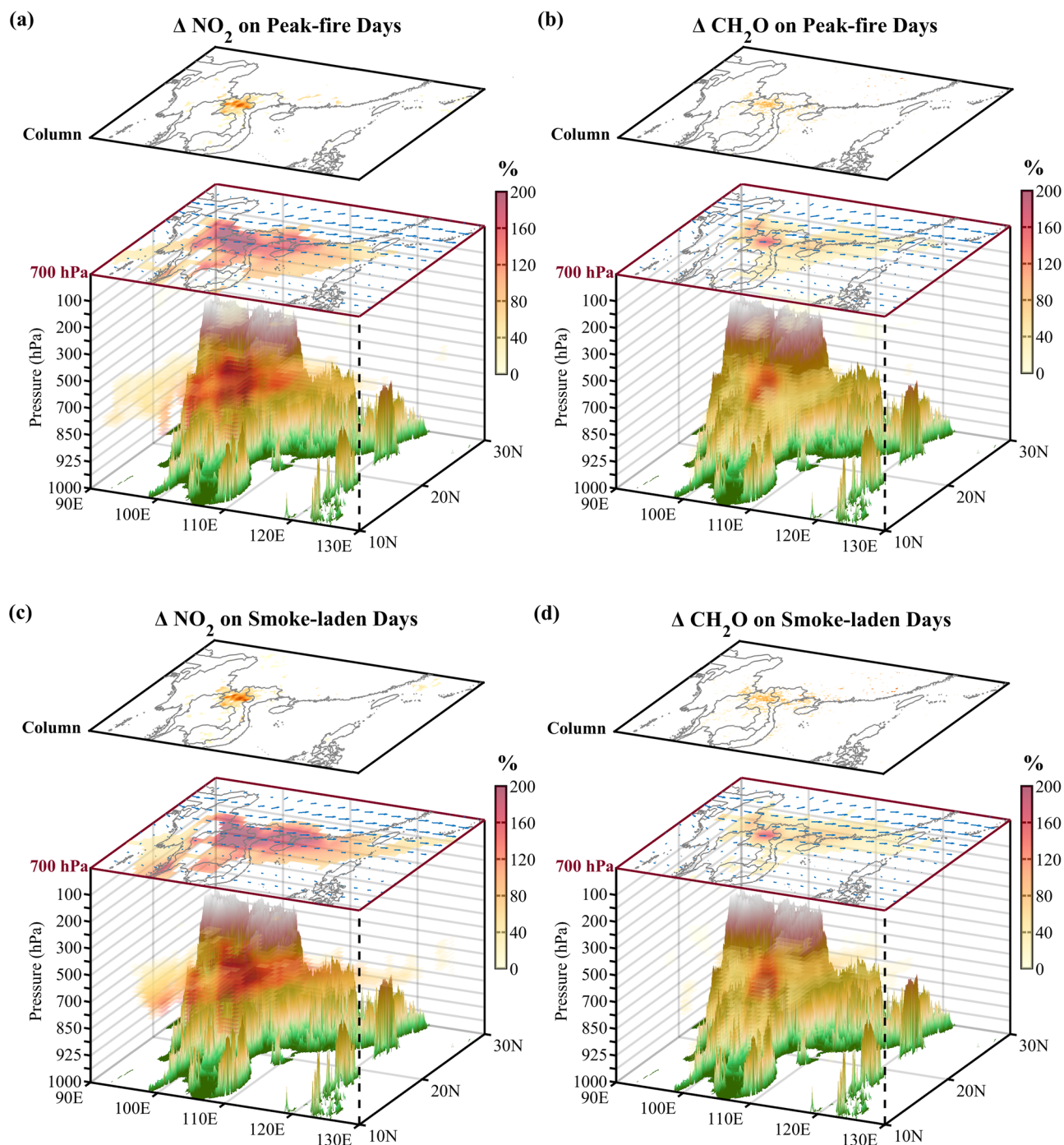


Fig. 3 | Excess O_3 precursor concentrations (ΔNO_2 and ΔCH_2O) relative to rare-fire days. Upper panels **a, b** represent the excess on peak-fire days, while lower panels **c, d** display the excess on smoke-laden days. Left panels **a, c** show the excess NO_2 concentrations (ΔNO_2), while right panels **b, d** represent the excess CH_2O concentrations (ΔCH_2O). Only areas with statistically significant increases at the 95%

confidence level are shown. Overlaid contours at the top of each panel denote the satellite-based excess of O_3 precursor column concentrations. Red-bordered contours highlight the excess of EAC4-based NO_2 and CH_2O concentrations and the fire-season wind fields at 700 hPa.

obscure the contribution of O_3 precursors from Indochina biomass burning to the atmospheric concentrations at the transport altitude when detected from the total tropospheric column. As shown by the EAC4 chemical reanalysis (Fig. 3 and Figure S5), the “terrain-chimney” effect is also pronounced for O_3 precursors. Significant enhancements of O_3 precursors are not confined to the near-source areas but extend downstream along the smoke transport altitude (e.g., 700 hPa), reaching as far east as the Bashi Strait. From the peak-fire days to the smoke-laden days (i.e., two days later), the high-enhancement region expands significantly eastward, indicating a

distinct transport signature of the smoke plumes. In the core transport corridor (from northern Laos to coastal South China), NO_2 increases at 700 hPa surpass 100%. In contrast, although CH_2O also shows a significant increase at this height, its enhancement is relatively smaller, mostly below 50%. Previous observational studies conducted in the downwind GBA region have shown that, relative to CH_2O , NO_2 concentrations decay more rapidly with altitude; even if surface NO_2 concentrations are higher than those of CH_2O , at higher altitudes, NO_2 concentrations become significantly lower than those of CH_2O ^{32,33}. Therefore, the long-range transport of

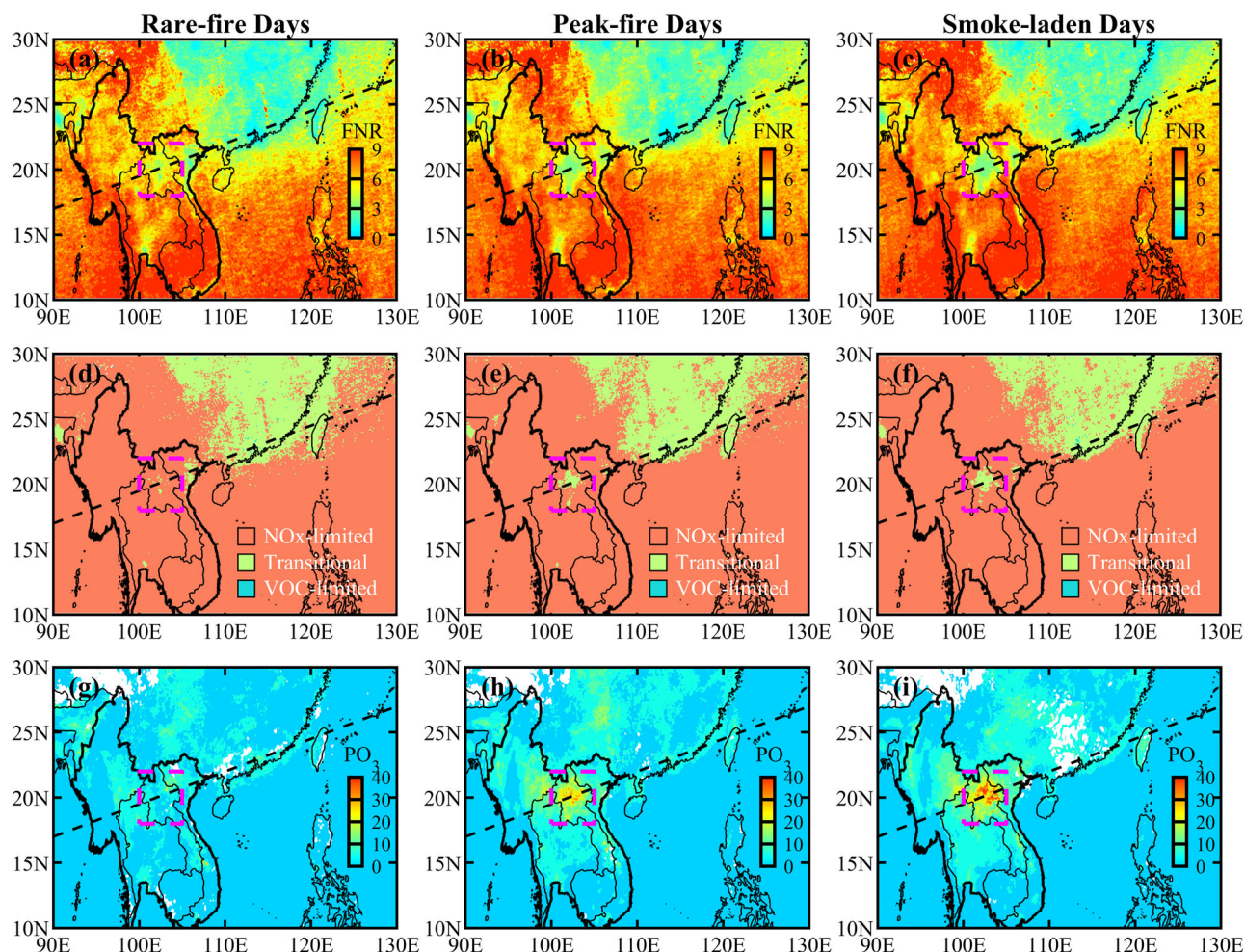


Fig. 4 | Spatial distributions of satellite-based O_3 formation chemistry indicators on rare-fire, peak-fire, and smoke-laden days. Upper panels a–c show the CH_2O/NO_2 ratio (FNR); middle panels d–f denote the O_3 formation sensitivity (OFS); and

lower panels g–i present the O_3 formation rate (PO_3 ; unit: $ppbv\ h^{-1}$). Magenta dashed boxes indicate northern Laos. Dashed lines mark the cross-sectional line used in Figs. 5 and 6.

Indochina biomass burning emissions within the lower free troposphere exerts a stronger perturbation on NO_2 than on CH_2O , which well explains the pronounced NO_2 enhancement observed downwind. This leads to the expectation that wildfire-borne NO_x will undergo complementary mixing with detained urban anthropogenic emissions in the downwind lower free troposphere, thereby enhancing photochemical O_3 production.

Efficient O_3 production chemistry within Indochina biomass burning plumes

In atmospheric chemistry, NO_x and VOCs are known to be the key precursors of O_3 production. However, the sensitivity of O_3 production to each precursor varies depending on various factors, particularly the relative availability of NO_x versus VOCs. Previous studies indicated that wildfire-related O_3 production evolves complex chemical processes, including alterations in O_3 formation sensitivity (OFS) and interactions between wildfire emissions and anthropogenic emissions^{13,14}. To better understand the O_3 production in Indochina biomass burning plumes, it is necessary to determine the OFS in both the burning source regions and the downwind receptor regions.

Satellite-derived CH_2O/NO_2 ratios (FNR) can provide regional photochemical information with good temporal and spatial continuity. Figure 4 shows the spatial distributions of satellite-based FNR, the corresponding OFS regime and the net O_3 formation rate (PO_3) on rare-fire, peak-fire, and smoke-laden days. In northern Laos, a marked increase in NO_2 column reduces FNR from 6.2 on rare-fire days to 5.5 on peak-fire and smoke-laden

days, and expands the transitional OFS regime from only 2.9% to 17–22% of the area, while other regions show no significant changes. On rare-fire days, the OFS regime across most of the Indochina Peninsula is NO_x -limited, indicating that O_3 production is primarily controlled by NO_x availability. Thus, large amounts of NO_x from biomass burning on peak-fire days can efficiently promote O_3 formation via the $NO_x \rightarrow O_3$ pathway. In northern Laos—a key biomass burning hotspot—nearly 20% of the region shifts to transitional OFS conditions, additionally activating the $VOC \rightarrow O_3$ pathway and allowing VOCs from biomass burning to further enhance O_3 production. Consistent with these changes in photochemical condition, the averaged PO_3 in northern Laos surges from 6.2 $ppbv\ h^{-1}$ on rare-fire days to 14.3 $ppbv\ h^{-1}$ on peak-fire days, and further reaches 17.6 $ppbv\ h^{-1}$ on smoke-laden days. These results provide robust evidence that biomass burning emissions lead to highly efficient O_3 production in the core burning region of northern Laos.

We further examine the three-dimensional characteristics of FNR and OFS based on EAC4 chemical reanalysis, which can avoid the masking effect of high anthropogenic emissions in downwind urban regions on the FNR and OFS changes caused by biomass burning emissions along the smoke transport aloft when viewed from a satellite-based column perspective. Figure 5 shows the vertical cross-section of EAC4-based FNR and OFS on rare-fire, peak-fire, and smoke-laden days. The cross-section passes through northern Laos—the core burning region—and the Greater Bay Area—the important anthropogenic emission region. Note that the OFS regime in Fig. 5 is identified based on the FNR thresholds derived from satellite-based

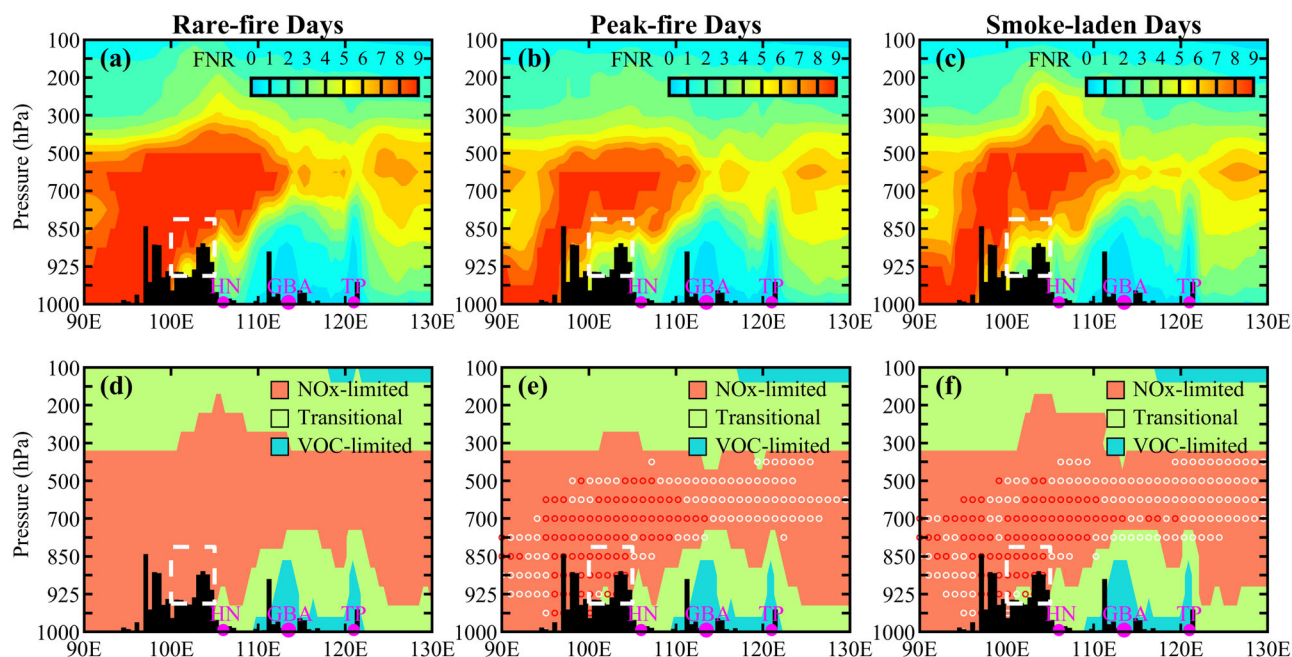


Fig. 5 | Cross-sections of EAC4-based O₃ formation chemistry indicators on rare-fire, peak-fire, and smoke-laden days. Upper panels (a–c) show the CH₂O/NO₂ ratio (FNR); middle panels (d–f) denote the O₃ formation sensitivity (OFS). Dots on peak-fire and smoke-laden days indicate locations with significant O₃ increases

relative to rare-fire days, while red dots additionally represent the significant NO₂ increases. HN, GBA, and TP denote Hanoi, the Greater Bay Area, and Taipei, respectively. White boxes denote the atmospheric boundary layer of northern Laos, and black blocks represent terrain elevation.

tropospheric columns. Thus, the selection of abovementioned cross-section facilitates validation of the EAC4-based OFS accuracy using the previously reported OFS observations in the Greater Bay Area (GBA). Based on the MAX-DOAS or UAV instruments in the GBA region, previous observations indicated that the VOC-limited regime dominated at the ground level, whereas the contribution of the NO_x-limited regime increased with altitude, becoming dominant above 0.8 km^{32–34}. Evidently, the EAC4-based OFS regime over the GBA region (Fig. 5) agrees well with these previous observations, justifying the applicability of satellite-based FNR thresholds to EAC4-based reanalysis data for identifying OFS regimes.

In Fig. 5, we observed that Indochina biomass burning emissions cause a significant decrease in FNR within both the atmospheric boundary layer over northern Indochina and the lower free troposphere in downwind regions. However, these changes in FNR do not lead to widespread shifts in the OFS regimes, except for a transition from NO_x-limited to the transitional regime in the lower boundary layer of northern Laos. Strictly speaking, from rare-fire days to peak-fire and smoke-laden days, the OFS remains consistently in the NO_x-limited regime across most of the smoke transport pathway. From a chemical perspective, this unchanged OFS regime strongly favors the conversion of additional NO_x emitted from biomass burning into elevated O₃ levels throughout the entire transport process via the NO_x → O₃ pathway²³. This favorable O₃ formation chemistry is distinctly different from that reported by Jin et al.³⁵ and Pouyaei et al.¹⁴, for North American wildfires. In North American, a larger increase in NO_x is observed only in NO_x-limited wildfire source regions, while a greater increase in CH₂O occurs in VOC-limited downwind urban areas, contributing to efficient O₃ production sequentially through the NO_x → O₃ and VOC → O₃ pathways. Evidently, due to the topographic configuration, the biomass burning plumes from northern Indochina are uplifted and have limited opportunity to intrude into the VOC-limited boundary layer in downwind urban regions, thus precluding the activation of the VOC → O₃ pathway. However, this does not imply that precursor emissions from Indochina biomass burning do not interact with anthropogenic emissions in downwind urban areas. Previous studies have shown that downwind urban areas with anthropogenic emissions establish a relatively high VOC background in the lower free troposphere³², which, when encountered the influx of pyrogenic

NO_x emissions, inevitably amplifies O₃ production via the NO_x → O₃ pathway. Compared with extra-tropical North America, the tropical Indochina Peninsula and its downstream regions feature both higher VOC background levels and stronger solar radiation. As a result, the additional injection of substantial NO_x from Indochina biomass burning into the NO_x-limited transport pathway acts upon these conditions, ultimately leading to more efficient photochemical O₃ production.

The above analysis indicates that the NO_x surge along the NO_x-sensitive transport pathway fuels efficient O₃ production in Indochina biomass burning plumes. Figure 6 presents the excess concentrations of O₃ (ΔO₃) and their normalized values (NEC_O₃) along the transport axis on peak-fire days and the following two days, relative to rare-fire days. Extremely high ΔO₃ (> 60 ppbv) remains confined in the airshed over northern Laos, indicating immediate and high-level O₃ production in the near-source areas. Subsequently, ΔO₃ decreases rapidly during the plume's transport over the Annamite Range due to the dilution effects, but recovers gradually after passing over Hanoi and eventually forms a secondary peak of ΔO₃ (~ 30 ppbv) over the GBA region. This recovery of ΔO₃ is consistent with ozonesonde observations at Hanoi (ΔO₃ = 28.9 ppbv, Fig. 1d) and Hong Kong (ΔO₃ = 32.2 ppbv, Fig. 1e), indicating the additional formation of O₃ resulting from the interaction between Indochina biomass burning emissions and downwind urban anthropogenic emissions during transport. In contrast to ΔO₃, NEC_O₃ allows us to examine the O₃ formation or loss processes without confounding influence of dilution. The NEC_O₃ concentrations increase rapidly after the smoke plume passes over Hanoi and then peak over the GBA region. On peak-fire days, the peak NEC_O₃ concentration exceeds 120 ppbv over the GBA region, which is nearly twice as high as the NEC_O₃ concentration over the biomass burning area. On the following two days, as the peak-fire emissions are transported downstream and mixed with detained urban anthropogenic emissions, the peak NEC_O₃ concentrations in the downwind region rise steadily day by day, eventually reaching values above 150 ppbv at more eastern locations over the GBA region on the smoke-laden days. These NEC_O₃ results suggest significant in-transit O₃ formation as Indochina biomass burning emissions interact with detained anthropogenic emissions in downwind urban areas, which well explains the secondary peak of ΔO₃ over the GBA region.

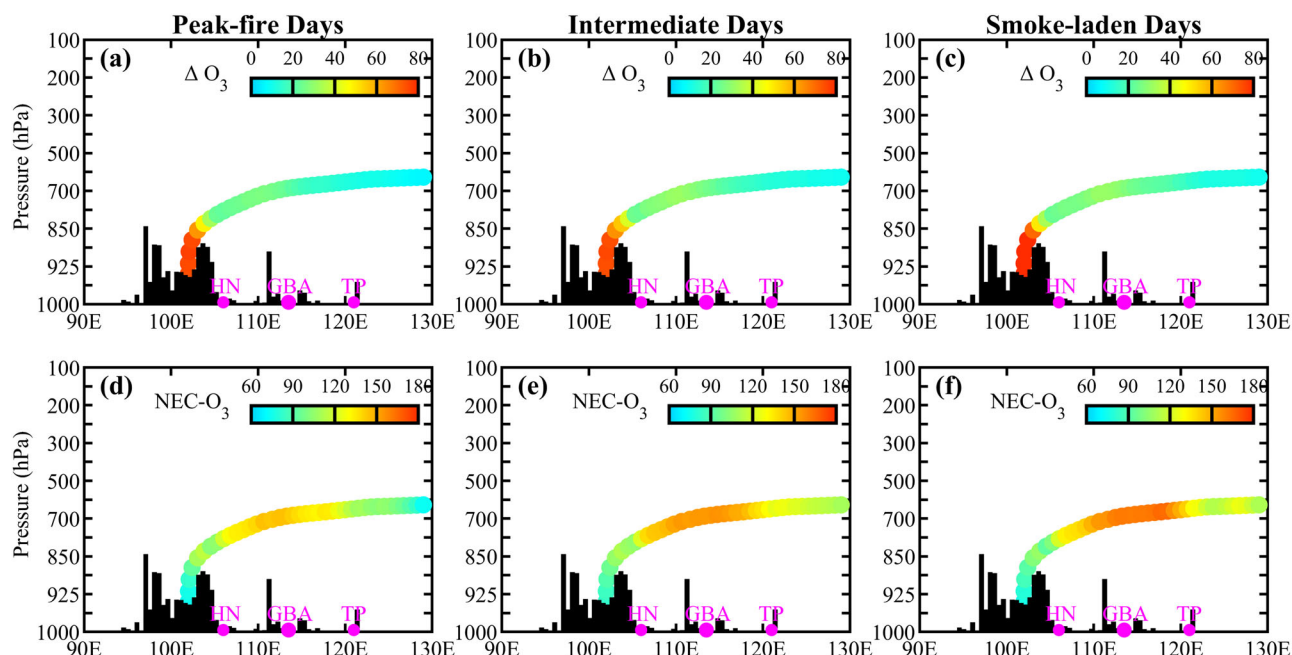


Fig. 6 | Excess O_3 concentrations (ΔO_3) and their dilution-corrected values (NEC- O_3) along the Indochina plume transport axis. Left, middle, and right panels show the ΔO_3 and NEC- O_3 on peak-fire (a, d), intermediate (b, e), and

smoke-laden days (c, f), respectively. HN, GBA, and TP denote Hanoi, the Greater Bay Area, and Taipei, respectively. Black blocks represent terrain elevation.

Overall, a combined analysis of ΔO_3 and NEC- O_3 along the smoke transport axis demonstrates the efficient O_3 production both over the burning area and in the downwind aloft. In the NO_x -sensitive downwind aloft, the maintenance of efficient O_3 production is contingent upon the NO_x enhancement driven by smoke plume transport and its synergistic mixing with detrained urban anthropogenic emissions. Given the multifaceted impacts of elevated O_3 levels on climate, air quality, and atmospheric chemistry, our findings underscore the urgency of implementing biomass burning prevention measures in the Indochina Peninsula.

Our study also has several limitations. First, the limited number of ozonesonde observations at Hanoi and Taipei somewhat complicates the robustness of cross-site comparisons. Nevertheless, multi-faceted comparisons, including comparisons between fire season and adjacent months, between smoke-laden and smoke-free days during fire season, and between ozonesonde measurements and EAC4 chemical reanalysis datasets, consistently corroborate the relative magnitude of LFT O_3 contributions from Indochina biomass burning across different ozonesonde sites. Second, the quantification of the Indochina biomass burning contribution to tropospheric O_3 enhancements is likely conservative, as it relies on a “rare-fire” baseline rather than a “no-fire” baseline. From an observational perspective, it is difficult to identify days with no biomass burning activity across the entire Indochina Peninsula during the fire season. Third, the selection of FNR threshold introduces potential uncertainties in OFS identification. We adopted a moderate FNR value of 4 as the upper bound of the transitional regime, and the derived OFS regimes show reasonable consistency with previous observations conducted in the GBA region. As a cautionary measure, we conducted two additional tests using commonly used and recently proposed upper bounds, i.e., 2 from Duncan et al.³⁶ and 6 from Jin et al.¹². Cross-comparisons reveal that the OFS regimes derived from all three FNR thresholds consistently indicate a NO_x -limited regime across most smoke transport pathways (Figures S7 and S9). This implies that the choice of the FNR threshold in this study does not fundamentally alter our core conclusions. Finally, the lack of validation of EAC4 against O_3 precursor observations may introduce some uncertainties in the characterization of O_3 -precursor chemistry. However, the EAC4 product has been well validated against multi-site ozonesonde observations, which may help

constrain its uncertainties. Despite these limitations, this work provided the first comprehensive assessment of O_3 production chemistry within Indochina biomass burning plumes, from near-source to downwind regions, offering useful insights for both local and regional O_3 mitigation strategies.

Methods

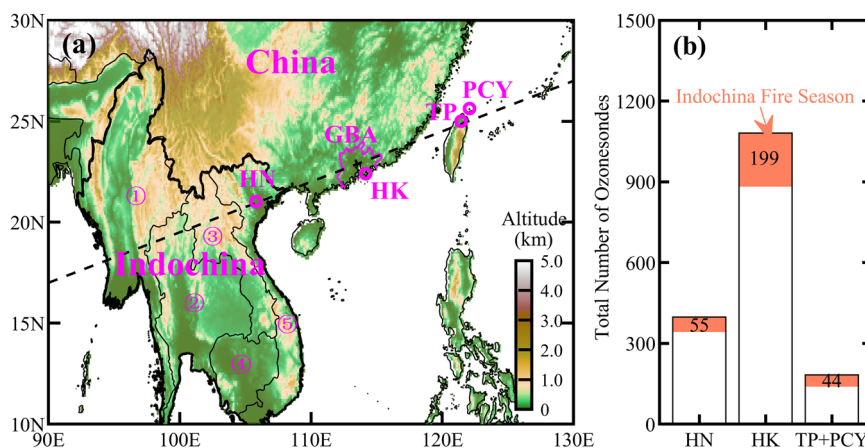
Ozonesonde dataset

The ozonesonde data used in this study comprise the publicly available observations from four downwind sites of the Indochina Peninsula—namely Hanoi, Hong Kong, Taipei, and Pengchiayu (Fig. 7a). The ozonesonde data at Hanoi are available in the SHADOZ (Southern Hemisphere Additional OZonesondes) archive <https://tropo.gsfc.nasa.gov/shadoz/>, while these at other three sites are available in the WOUDC (World Ozone and Ultraviolet Radiation Data Centre) archive <https://woudc.org/en/>. These ozonesondes are conducted at 14:00 local time, weekly at Hong Kong, and once or twice a month at other sites. Nearly all of the aforementioned sites have captured individual cases of tropospheric O_3 enhancement induced by Indochina biomass burning using ozonesonde observations^{29,37,38}. Figure 7b provides a summary of the ozonesonde observations at these four sites during the study period. Hong Kong, located in the Greater Bay Area (GBA) of South China, exhibits the highest ozonesonde frequency, with a total of 1,081 soundings, 199 of which were conducted during the Indochina fire season (March–April). In contrast, ozonesondes at Taipei and Pengchiayu are relatively sparse in time, with a total of 183 soundings. Given their geographical proximity, ozonesonde data from these two stations are combined for subsequent analyses.

Satellite remote sensing dataset

The used satellite remote sensing data include daily fire hotspots from the MODIS (Moderate Resolution Imaging Spectroradiometer) Collection 6 thermal anomaly products³⁹, tropospheric NO_2 and CH_2O column from the OMI (Ozone Monitoring Instrument) level-3 products and the TROPOMI (Tropospheric Monitoring Instrument) level-2 products, and ozone production rate (PO₃) estimated from OMI and TROPOMI observation^{40,41}. For fire products, we exclude these data prior to 2003, accounting for the Terra (launched December 1999) and Aqua (May 2002) satellite

Fig. 7 | Study region, site distribution and sample size of ozonesondes. (a) Study area and ozonesonde sites. Numbers ①–⑤ denote Myanmar, Thailand, Laos, Cambodia, and Vietnam, respectively. HN, HK, TP, and PCY represent the ozonesonde sites at Hanoi, Hong Kong, Taipei, and Pengchiayu, respectively. GBA denotes the Guangdong-Hong Kong-Macau Greater Bay Area. The dashed line shows the cross-sectional line used in this study. (b) Total number of ozonesondes at different sites during the study period. Orange bars highlight the number of ozonesondes during the Indochina fire season (March–April).



commissioning periods. For OMI/TROPOMI products, both NO_2 and CH_2O columns were pre-filtered with an effective cloud fraction $> 30\%$ and solar zenith angles of $> 70\%$ (for CH_2O) and $> 85\%$ (for NO_2). In this study, we additionally remove these data with data-quality flags not designated as good, as well as these unrealistically large positive and negative values following Zhu et al.⁴² (for CH_2O) and Johnson et al.³⁰ (for NO_2). Prior to further analysis, all the above satellite datasets were spatially re-gridded to a consistent $0.1^\circ \times 0.1^\circ$ resolution.

A spatiotemporal fusion framework is applied to integrate the OMI and TROPOMI products. We first extract the overlapping observational data from the two sensors and examine their consistency. For variables that exhibit pronounced biases—and given that TROPOMI offers superior data quality relative to OMI^{31,43}—we use the overlapping TROPOMI observations as a baseline to derive linear regression functions relating the two products⁴⁴. Once the linear regression relationship is established, we correct all OMI data, including those outside the overlapping period. Then, we integrate the TROPOMI observation into the corrected OMI data by taking the mean value at overlapping grid cells and inserting the TROPOMI value at non-overlapping grid cells, thereby accomplishing cross-satellite data fusion. Figure S9 shows the consistency of overlapping original data between the OMI and TROPOMI products, as well as the data consistency after the OMI correction. The correction procedure significantly enhances the data consistency between the two satellite products, particularly for large values. Note that we ultimately did not apply the correction to the OMI CH_2O data, because its cumulative probability density is almost consistent with that of the TROPOMI products.

EAC4 chemical reanalysis

The Copernicus Atmosphere Monitoring Service's Atmospheric Composition Reanalysis 4 (EAC4), the fourth generation ECMWF global reanalysis of atmospheric composition, is used to characterize the three-dimensional O_3 and its precursors (NO_2 and CH_2O) within Indochina biomass burning plumes. EAC4 adopts an extended version of the Carbon Bond 2005 chemical mechanism with anthropogenic and natural emission inputs, and also assimilates satellite observations of O_3 , NO_2 , and other species with a 4D-Var method⁴⁵. The EAC4 Global Fire Assimilation System utilizes satellite observations of fire radiative power to estimate the pyrogenic emissions from biomass burning and vegetation fires⁴⁶. To evaluate whether EAC4 can accurately represent atmospheric composition in Indochina biomass burning plumes, tropospheric O_3 concentrations from EAC4 products (06:00 UTC) are compared to fire-season ozonesonde measurements at Hanoi, Hong Kong, Taipei and Pengchiayu (Figure. S3). The validation results confirm an overall high correlation ($R = 0.76$) between EAC4 reanalysis and ozonesonde observations in the troposphere, with a mean absolute bias (MAB) below 13 ppbv. The significant overestimation of O_3 by EAC4 in the atmospheric boundary layer is most likely attributable to model

deficiencies in representing loss processes and emission strength. However, in the lower free troposphere, where long-range transport of Indochina biomass burning plumes occurs, EAC4 shows well agreement with ozonesondes ($R = 0.69$, $\text{MAB} = 10.63$ ppbv). Overall, these validations demonstrated the capability of EAC4 products to capture the free-tropospheric O_3 variability during the Indochina fire season.

Identification of ozone formation sensitivity

Tropospheric O_3 formation exhibits a nonlinear response to its precursors and can be broadly categorized into three sensitivity regimes: VOC-limited, NO_x -limited and transitional^{35,36}. We diagnose ozone formation sensitivity (OFS) using the ambient concentration ratio of two indicator species— CH_2O to NO_2 (hereafter referred to as FNR)—which serves as a proxy for VOCs and NO_x directly involved in O_3 - NO_x -VOC chemistry. The FNR is a widely adopted indicator of OFS in both satellite-based and model-based diagnostics^{30,35,47–50}. Early studies suggested that when FNR exceeds 2, ozone formation is limited by NO_x availability; when FNR falls below 1, it is limited by VOCs; and values between 1 and 2 indicate sensitivity to both precursors³⁶. However, recent studies have shown that the upper bound of the transitional regime can be as high as 4^{35,51}, and occasionally exceeds this value (e.g., reaching 6) within wildfire plumes¹². Given the lack of available surface O_3 observations over the Indochina Peninsula, we are unable to establish local FNR thresholds for this biomass burning study. We therefore use the moderate FNR thresholds 1–4 from these previous studies for our OFS identification, and also test alternative thresholds (1–2 and 1–6) to evaluate the sensitivity of our results (Figures. S7 and S8).

Calculation of normalized excess O_3 concentration

In this study, the normalized excess concentration (NEC) is also calculated for EAC4-based O_3 on peak-fire (smoke-laden) days, relative to rare-fire days. Given that carbon monoxide (CO) is a relatively stable component with low reactivity, many previous biomass burning studies have normalized other trace gas concentrations to CO to investigate loss or formation processes without confounding influence of dilution within the transported biomass burning plume^{52–54}. By definition, NEC is the excess concentration of species X (ΔX) in the smoke plume, normalized by the excess concentration of CO at the same position (ΔCO) and that at the burning source ($\Delta \text{CO}_{\text{source}}$), as shown in the following equation⁵²:

$$\text{NEC}_X = \Delta X \frac{\Delta \text{CO}_{\text{source}}}{\Delta \text{CO}} \quad (1)$$

Here, we identified smoke plume on peak-fire (smoke-laden) days in a three-dimensional space, characterized by significant increases in EAC4-based O_3 or CO concentrations at the 95% confidence level relative to rare-fire days. Subsequently, we determined the most intense burning source at

the location associated with the maximum ΔCO value on peak-fire days, thereby yielding a $\Delta\text{CO}_{\text{source}}$ value relative to rare-fire days. Finally, this $\Delta\text{CO}_{\text{source}}$ value was used to calculate the normalized excess concentration of O_3 (NEC_{O_3}) in transported smoke plumes. It should be noted that extremely low ΔCO values at the plume edges can introduce large uncertainties in NEC_{O_3} , therefore, this study retained only the NEC_{O_3} values along the plume axis for analysis.

Other supporting methodology

This study involved extensive comparative statistics between smoke-free (rare-fire) and smoke-laden (peak-fire) days, for which the statistical significance of the differences was assessed using the Student's t-test. To explore smoke transport, we conducted air mass trajectory analysis using the HYSPLIT model with NCEP/NCAR reanalysis inputs⁵⁵, calculating 96-hour backward trajectories ending at 3.5 km altitude (the median height of the LFT as defined in this study) above Hong Kong. To verify our receptor-view classification, we calculated the trajectory-fire interception (TFI) following Schneider et al.²⁷. Briefly, the TFI method spatially compares 96-hour backward trajectories with the locations of biomass burning fire hotspots detected by the MODIS satellite-based instruments. For a single trajectory point, a TFI is identified if a trajectory point and a fire hotspot are within 0.5° of each other. The total number of TFIs is then integrated for each complete trajectory. Besides, we adopted the ERA5 meteorological reanalysis (06:00 UTC) to trace the changes of wind field and boundary layer height. (Figs. 6 and 7).

Data Availability

All data used in this study are publicly available according to the following archives (last access date: 31 May 2026): Ozone sonde at Hanoi: <https://tropo.gsfc.nasa.gov/shadoz/>; Ozone sonde at Hong Kong, Taipei, and Pengchiayu: <https://woudc.org/en/>; OMI- and TROPOMI-based O_3 precursor column: <https://disc.gsfc.nasa.gov/>; OMI- and TROPOMI-based O_3 production rate: <https://www.ozonrates.space/>; EAC4-based O_3 and its precursors reanalysis: <https://ads.atmosphere.copernicus.eu/datasets/cams-global-reanalysis-eac4?tab=overview>; ERA5 meteorological reanalysis: <https://cds.climate.copernicus.eu/>; Global biomass burning emissions: <https://ads.atmosphere.copernicus.eu/datasets/cams-global-fire-emissions-gfas?tab=overview>.

Code availability

The code generated and/or analyzed during the current study are not publicly available due to the large volume of input data but are available from the corresponding author on reasonable request.

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Author contributions

Z.L. conceived the original idea, analyzed the data, and wrote the manuscript. F.L. analyzed the biomass burning emission and satellite data. J. X analyzed the meteorological and chemical reanalysis data. Z.M. supervised the research project. All authors discussed the results and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Zhiqiang Ma.

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